

INTERNSHIP PROJECT REPORT

Predictive Shipment Delay Risk Intelligence System

Organization:

APL Logistics (KWE Group)

Project Domain:

Supply Chain Analytics & Machine Learning

Dataset:

APL_Logistics.xlsx | 180,519 Orders | 40 Features

Mentor:

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1. Executive Summary

This internship project, undertaken with APL Logistics (a member of the KWE Group), addresses one of the most pressing operational challenges in modern global logistics: predicting shipment delays before they occur. Drawing on a dataset of 180,519 orders spanning five global markets, the project builds an end-to-end machine learning pipeline that assigns each order a quantified Late Delivery Probability and a Risk Category — enabling operations teams to act proactively rather than reactively.

Key Finding: 54.8 % of all orders in the dataset carry a high late-delivery risk (Late_delivery_risk = 1), making predictive intervention a business-critical capability rather than a nice-to-have.

The project follows the complete Data Science methodology — from raw data exploration and feature engineering, through model development and evaluation, to a deployable Streamlit risk dashboard. Outputs include risk probability scores, explainable feature drivers, and a prioritised action queue for logistics planners.

2. Organization Background

2.1 About APL Logistics

APL Logistics is a global third-party logistics (3PL) provider operating under the KWE (Kintetsu World Express) Group. The company manages end-to-end supply chain solutions across warehousing, freight management, and distribution networks spanning Latin America, Europe, Pacific Asia, USCA (US, Canada & Australia), and Africa.

2.2 Business Context

- Serves Consumer, Corporate, and Home Office customer segments across 180,000+ annual orders.
- Operates across multiple shipping modes: Standard Class, Second Class, First Class, and Same Day.
- Products span 10+ categories including Cleats, Men's Footwear, Women's Apparel, Outdoor Games, and Electronics.
- Late deliveries directly trigger SLA breaches, financial penalties, and customer churn.

3. Problem Statement

3.1 Core Challenge

APL Logistics lacks a forward-looking system to identify high-risk orders before they are shipped. Current operations are reactive — delays are handled after they occur, resulting in:

- Last-minute emergency re-routing costs
- SLA penalty payments to customers

- Inability to prioritise high-value delayed orders
- No quantitative risk scoring for operational planning

3.2 Research Question

Which orders are likely to be delayed — and why — before they are shipped?

3.3 Objectives

- Develop a binary classification model predicting Late_delivery_risk (1 = delayed, 0 = on time).
- Assign each order a Late Delivery Probability (0–1) and a Risk Category (Low / Medium / High).
- Identify and explain the key drivers of delivery risk.
- Build a Streamlit dashboard for real-time operational use.

4. Dataset Overview & Exploratory Analysis

4.1 Dataset Summary

Parameter	Value
Total Records	180,519
Number of Features	40
Target Variable	Late_delivery_risk (Binary: 0 / 1)
Time Period	Multi-year order history
Markets Covered	LATAM, Europe, Pacific Asia, USCA, Africa
Shipping Modes	Standard Class, Second Class, First Class, Same Day

4.2 Target Variable Distribution

The dataset exhibits a moderate class imbalance with late delivery orders forming the majority class:

Delivery Risk	Count	Percentage
High Risk (Late Delivery = 1)	98,977	54.8 %
Low Risk (On Time = 0)	81,542	45.2 %
Total	180,519	100.0 %

4.3 Delivery Status Breakdown

Delivery Status	Count	% of Total
Late Delivery	98,977	54.8 %
Advance Shipping	41,592	23.0 %
Shipping On Time	32,196	17.8 %
Shipping Canceled	7,754	4.3 %

4.4 Market & Shipping Mode Distribution

Market	Orders	Late Risk Rate
LATAM	51,594	54.4 %
Europe	50,252	55.2 %
Pacific Asia	41,260	55.0 %
USCA	25,799	54.8 %
Africa	11,614	54.6 %

Shipping Mode	Orders	Late Risk Rate
Standard Class	107,752	38.1 %
Second Class	35,216	76.6 %
First Class	27,814	95.3 %
Same Day	9,737	45.7 %

Critical Insight: First Class shipping carries a 95.3 % late delivery risk rate — the highest of any mode — suggesting systemic capacity or routing failures for express shipments. This is a primary driver for feature engineering.

4.5 Order Financial Metrics

Financial Metric	Mean	Std Dev	Min	Max
Sales per Order (\$)	203.77	249.64	0.00	9,999.99

Benefit per Order (\$)	21.97	97.12	-2,137.49	913.89
Order Profit per Order (\$)	21.97	104.43	-4,274.98	911.80
Order Item Product Price (\$)	—	—	9.99	999.99

4.6 Shipping Days Analysis

Metric	Real Shipping Days	Scheduled Shipping Days
Mean	3.50	2.93
Median	3.00	4.00
Std Dev	1.62	1.37
Minimum	0	0
Maximum	6	4

The mean real shipping duration (3.50 days) exceeds the mean scheduled duration (2.93 days), confirming a systemic pattern of delays across the logistics network.

5. Data Science Methodology

5.1 Data Preprocessing

Missing & Inconsistent Values

- Identified and imputed missing values using median imputation for numerical fields and mode imputation for categorical fields.
- Removed duplicate order entries and standardised inconsistent country/city name formats (e.g., 'EE. UU.' → 'United States').

Feature Encoding

- One-Hot Encoding applied to nominal categorical variables: Shipping Mode, Market, Customer Segment, Order Status.
- Target Encoding applied to high-cardinality columns: Category Name, Department Name, Order Region.

Feature Scaling

- StandardScaler applied to continuous numerical features for Logistic Regression compatibility.
- Tree-based models (Random Forest, XGBoost) trained on unscaled features as they are scale-invariant.

Class Imbalance

- ▶ Dataset exhibits a 54.8 / 45.2 class split — moderate imbalance addressed via `class_weight='balanced'` in model training.
- ▶ SMOTE (Synthetic Minority Oversampling Technique) applied as an alternative strategy and compared against weighted training.

5.2 Feature Engineering

Beyond raw dataset fields, the following engineered features were created to improve model predictive power:

Engineered Feature	Formula / Logic	Business Rationale
Shipping Pressure Index	Scheduled Days / Order Quantity	Captures per-unit time pressure
Delay Buffer	Real Shipping Days – Scheduled Days	Directly measures schedule adherence
Express Mode Flag	1 if First Class or Same Day	Flags high-risk shipping modes
Order Complexity Score	Quantity × Product Price	Proxy for order handling difficulty
Discount Pressure Flag	1 if Discount Rate > 15 %	High-discount orders may be deprioritised
Profit Risk Flag	1 if Profit Ratio < 0	Loss-making orders — fulfillment risk

5.3 Model Development

Models Trained

- ▶ Logistic Regression — interpretability baseline, establishes minimum performance threshold.
- ▶ Random Forest Classifier — handles non-linear relationships, robust to noise.
- ▶ Gradient Boosting / XGBoost — ensemble method capturing complex feature interactions.

Train-Test Split

- ▶ 80 % training / 20 % test split with stratified sampling to preserve class distribution.
- ▶ Cross-validation (5-fold) used to validate model stability.

5.4 Model Evaluation Metrics

Metric	Business Meaning	Priority
ROC-AUC	Overall discrimination quality between risk classes	Primary
Recall (Sensitivity)	Ability to catch true delay cases — reduces missed alarms	High

Precision	Avoidance of false delay alarms — reduces unnecessary interventions	Medium
F1 Score	Balanced measure of precision and recall	High
Confusion Matrix	Breakdown of TP, TN, FP, FN for operational planning	Diagnostic

5.5 Risk Probability & Classification

Each order receives two outputs from the model:

- ▶ Late Delivery Probability: A continuous score from 0.0 to 1.0 produced by `model.predict_proba()`.
- ▶ Risk Category: Derived from the probability score using operational thresholds.

Risk Category	Probability Threshold	Recommended Action
Low Risk	0.0 – 0.4	Standard processing; no intervention required
Medium Risk	0.4 – 0.7	Monitor order; flag for daily review
High Risk	0.7 – 1.0	Immediate intervention; reroute or notify customer

6. Key Findings & Insights

6.1 Top Delay Risk Drivers

Feature importance analysis from the Random Forest and XGBoost models identifies the following as the strongest predictors of late delivery:

Rank	Feature	Importance	Insight
1	Shipping Mode (First Class)	Very High	95.3 % late risk — express channel is failing
2	Days for Shipping (Real)	High	Actual duration directly predicts delay
3	Delay Buffer	High	Positive buffer = late delivery confirmed
4	Days for Shipment (Scheduled)	High	Tight schedules correlate with delays
5	Order Item Discount Rate	Medium	Heavy discounting predicts fulfillment issues
6	Order Item Profit Ratio	Medium	Negative-margin orders are delay-prone
7	Market (Europe)	Medium	Slightly elevated regional risk
8	Order Quantity	Low-Medium	Higher quantity = more handling complexity

6.2 Regional Findings

- ▶ All five global markets (LATAM, Europe, Pacific Asia, USCA, Africa) show remarkably uniform late delivery risk rates between 54.4 % and 55.2 %, indicating that delay risk is driven by order-level and shipping mode factors rather than geography.
- ▶ Central America (28,341 orders) and Western Europe (27,109 orders) are the two highest-volume order regions, making them priority targets for intervention programs.

6.3 Customer Segment Findings

- ▶ Consumer (54.8 %), Corporate (54.7 %), and Home Office (55.1 %) segments all exhibit similar delay risk — risk management must apply uniformly across segments.
- ▶ Home Office, despite being the smallest segment, carries the marginally highest risk and may warrant dedicated SLA monitoring.

6.4 Financial Impact

- ▶ Average benefit per order: \$21.97 — with high variance (std dev \$97.12), delayed high-value orders represent significant financial exposure.
- ▶ Orders with negative profit ratios show elevated delay risk, suggesting that loss-making fulfillment is a systemic operational signal.
- ▶ Discount Rate above 15 % correlates with higher delay probability — deep promotions may outpace logistics capacity.

7. Streamlit Dashboard Design

7.1 Dashboard Architecture

The operational risk dashboard is built in Python using Streamlit and is structured into four functional modules:

Module	Purpose	Key Components
Delay Risk Overview	Portfolio-level risk snapshot	Overall risk distribution chart, High-risk order count, Daily/weekly trend
Order-Level Prediction	Individual order risk scoring	Probability score (0–1), Risk category badge, Top contributing factors
Region & Mode Analysis	Spatial and modal risk heatmaps	Risk by market/region, Shipping mode comparison, Interactive filters
Operations Action Panel	Prioritised intervention queue	High-risk order list, Risk-based sorting, Export to CSV

7.2 User Interaction Controls

- ▶ Shipping Mode Filter — isolate risk view by Standard / Second / First / Same Day.
- ▶ Market / Region Selector — drill into specific geographic zones.

- ▶ Risk Threshold Slider — adjust High/Medium/Low boundaries dynamically.
- ▶ Customer Segment Filter — focus on Consumer, Corporate, or Home Office.

7.3 Technical Stack

Component	Technology
Web Framework	Streamlit (Python)
ML Models	scikit-learn, XGBoost
Data Processing	pandas, NumPy
Visualisation	Plotly, Matplotlib, Seaborn
Explainability	SHAP (SHapley Additive exPlanations)
Deployment	Streamlit Cloud / Docker

8. Key Performance Indicators (KPIs)

KPI	Description	Target / Benchmark
Late Delivery Probability	Per-order likelihood score (0–1)	Model Calibration AUC > 0.80
Risk Category Accuracy	% of high-risk orders correctly flagged	Recall > 85 %
False Alarm Rate	% of on-time orders incorrectly flagged as high-risk	Precision > 75 %
High-Risk Order Volume	Count of orders requiring immediate intervention	Operational dashboard KPI
Intervention Success Rate	% of flagged orders that avoid delay post-intervention	Operations tracking KPI
Model Drift Rate	Degradation in AUC over rolling 30-day window	< 3 % monthly drift

9. Recommendations

9.1 Immediate Operational Actions

- ▶ Conduct a root-cause audit of First Class shipping — with a 95.3 % delay rate, the express routing infrastructure requires urgent capacity and process review.
- ▶ Implement the Streamlit dashboard in the operations center as a daily briefing tool for logistics coordinators.
- ▶ Set automated alerts for any order where predicted delay probability exceeds 0.70 on the day of order processing.

9.2 Strategic Improvements

- ▶ Deploy SHAP-based explainability in customer communication: proactively notify customers when their order is flagged as High Risk before shipping.
- ▶ Integrate discount management with logistics capacity planning to prevent high-promotion periods from overwhelming shipping capacity.
- ▶ Establish a model retraining pipeline (monthly) to prevent prediction drift as logistics patterns evolve.

9.3 Data & Technology Roadmap

- ▶ Enrich the dataset with real-time weather, port congestion, and carrier capacity data to improve prediction accuracy.
- ▶ Build an API endpoint wrapping the ML model so downstream ERP and order management systems can consume risk scores at order creation time.
- ▶ Explore deep learning sequence models (LSTM) for multi-step delay forecasting as order volume scales.

10. Conclusion

This internship project successfully transforms APL Logistics' supply chain operations from a reactive delay management model to a predictive risk intelligence capability. By analysing 180,519 historical orders across five global markets, the project demonstrates that delivery risk can be reliably predicted using a combination of shipping mode, order characteristics, and engineered timing features.

The 54.8 % baseline late delivery rate in the current data underscores the urgent need for this capability. The machine learning models developed — Logistic Regression, Random Forest, and XGBoost — provide both predictive accuracy and operational explainability, ensuring logistics teams can trust and act on the risk scores generated.

The Streamlit dashboard translates complex model outputs into a clear, actionable interface for logistics coordinators, enabling risk-based order prioritisation, targeted interventions, and proactive customer communication. Together, these outputs form the predictive backbone of an intelligent, data-driven supply chain at APL Logistics.

This project bridges the gap between historical performance analysis and forward-looking operational intelligence — enabling APL Logistics to act before delays happen, not after.

References & Data Sources

- ▶ APL Logistics Order Dataset — APL_Logistics.xlsx (180,519 records, 40 features)
- ▶ Unified Mentor Project Brief — Predictive Shipment Delay Risk Intelligence
- ▶ scikit-learn Documentation — <https://scikit-learn.org/>

- ▶ XGBoost Documentation — <https://xgboost.readthedocs.io/>
- ▶ SHAP (SHapley Additive exPlanations) — Lundberg & Lee, 2017
- ▶ Streamlit Documentation — <https://docs.streamlit.io/>
- ▶ SMOTE: Chawla et al. (2002) — Synthetic Minority Oversampling Technique